GAYATRI GUNJAL

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CAREER OBJECTIVE

To utilize my technical knowledge in fulfilling my career and organization objectives as well as to synergize my efforts with the organization's vision to work in an innovative and competitive environment.

EDUCATION

NDMVP K. B. T. College of Engineering, Nashik [CGPA: 8.52]
Bachelor of Engineering in Electronics and Telecommunication Engineering — 2019 — 2022

NDMVP R. S. M. Polytechnic, Nashik [Percentage: 75.12%]
Department of Electronics and Telecommunication — 2016 — 2019

C. D. O. Meri Highschool, Nashik [Percentage: 82.80]
Secondary School Leaving Certificate (SSLC) — 2015 — 2016

WORK EXPERIENCE

Embedded Developer Trainee

Emertxe Information Technologies — 09/2023 – 07/2024

Currently, I have successfully completed the hands-on technical training program — Emertxe Certified Embedded Professional (ECEP) (This course is Government of India certified program, aligned with Skill India / NSDC under Electronics Sector Skill Council of India)

TECHINCAL SKILLS

Programming Languages:
- Advanced C Programming
- Data structures & Algorithm
- Shell Scripting
- OOP using C++

Embedded controllers:
- Memory
- Interfacing, Character LCD
- Peripherals usage – Timers, Counters
- Communication protocols - UART, SPI, I2C, CAN

Embedded platforms:
- Distributions - Linux (Ubuntu)
- PIC (PIC16F877A) board

Development environment and tools:
- Dev environment: Vim, VScode
- Compilers: GCC, XC8

PROJTECTS

Project Number:1

Title	Image Steganography using LSB Encoding and Decoding
Project brief	The objective was to send a secret text file encoded inside an image of bmp file format. Encoded the length of the secret text and then encoded the data into the LSB of the image bytes. The decoding process involves decoding the length and then decoding the text bit by bit. The final output is the secret text after decoding.
Technologies used	Embedded C — File operations, Pointers, Bitwise operations, Functions, Command line arguments

Project Number:2

Title	Inverted Search
Project brief	An inverted index is an index data structure storing a mapping from content, such as words or numbers, to its locations in a database file, or in a document or a set of documents. The purpose of an inverted index is to allow fast full text searches, at a cost of increased processing when a document is added to the database. The inverted file may be the database file itself, rather than its index. It is the most popular data structure used in document retrieval systems, used on a large scale for example in search engines. The purpose of this project is to implement the inverted search using Hash Algorithms.
Technologies used	Embedded C — File operations, Pointers, Bitwise operations, Functions, Command line arguments

Project Number:3

Title	Car Black Box
Project brief	Black Boxes are typically used in any transportation system (ex: Airplanes) that are used for analysis post-crash and understand the root cause of accidents. Continuous monitoring and logging of events (ex: over-speeding) is critical for effective usage of black box. The goal of this project is to implement core functionalities of a car black-box in a PIC based micro-controller supported by rich peripherals. Events will be logged in EEPROM in this project. This project can be further extended to any vehicle.
Technologies used	Embedded C — PIC(16F877A) board, Character LCD, Matrix Key Pad, External eeprom, RTC, Timers